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The control of isariopsis leaf spot and downy mildew in grapevine cv. Isabel with the essential oil of lemon grass and the activity of defensive enzymes in response to the essential oil

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ABSTRACT

The aim of this study was to evaluate the potential of the essential oil of lemon grass (*Cymbopogon citratus*) for the control of isariopsis leaf spot (*Pseudocercospora vitis*) and downy mildew (*Plasmopara viticola*) in grapevine cv. Isabel, the effect of the essential oil on the productivity of the grapevines and the effect of the essential oil on the activity of the enzymes chitinase and catalase. The experiment was conducted in a commercial vineyard over two consecutive crop cycles. Each plant in the experiment was subjected to one of the following nine treatments: 0, 0.5, 1.0, 2.0 or 4.0 mL L^{-1} essential oil, Tween® 80%, Bordeaux mixture, Acibenzolar-S-methyl and mancozeb. An analysis of the area under the disease progress curve showed a quadratic response by both diseases to the doses of essential oil during the first and second crop cycles. The essential oil treatments also increased the number and mass of the clusters of fruit as well as the productivity and desirable chemical characteristics of the grape. The activity of chitinase increased as a result of the essential oil treatments, whereas the activity of catalase decreased. The essential oil at doses of 1.0 and 2.0 mL L^{-1} can serve as an alternative means of controlling isariopsis leaf spot and downy mildew and can also serve to improve the fruit quality of grapes cv. Isabel in tropical regions.

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1. Introduction

Most cultivated grapevines are susceptible to various diseases, such as downy mildew (*Plasmopara viticola* (Berk. & M.A. Curtis) Berl. & De), anthracnose (*Elsinoe ampelina* Shear), isariopsis leaf spot (*Pseudocercospora vitis* (Lév.) Speg.), gray rot (*Botrytis cinerea* Pers) and ripe rot (*Glomerella cingulata* (Stoneman) Spauld & Schrenk). Thus, control methods for these diseases are necessary and of particular importance because they need to be effective and efficient and should still result in a competitive production cost in the market (Farjado, 2003).

The most widely used fungicide to control diseases of grapevine is Bordeaux mixture, a copper fungicide. However, like all copper compounds, this fungicide may produce symptoms of toxicity in young plant tissues, and the corrosive action of the fungicide may compromise the structure of the vine. Due to these characteristics, Bordeaux mixture is recommended for use only after fruiting. Other synthetic fungicides, such as metalaxyl, thiophanate-methyl and cymoxanil, are used on a large scale in the wine industry for the control of these diseases (Amorim and Kuniyuki, 2005).

The use of fungicides can increase production significantly. However, fungicide use can also cause serious damage to the environment and can result in the selection of pathogens that are resistant to certain active ingredients contained in the fungicides. Consumers of fresh fruit are increasingly concerned about the origin of the products that they purchase, the presence of toxic waste and the conservation of the fruit. For this reason, the consumption of organic products has increased steadily, causing changes in the production, storage, distribution and marketing of agricultural products (Flores-Cantillano et al., 2001). This ecological awareness has created opportunities for agriculture, enhancing trade in organic products and leading some growers to shift from

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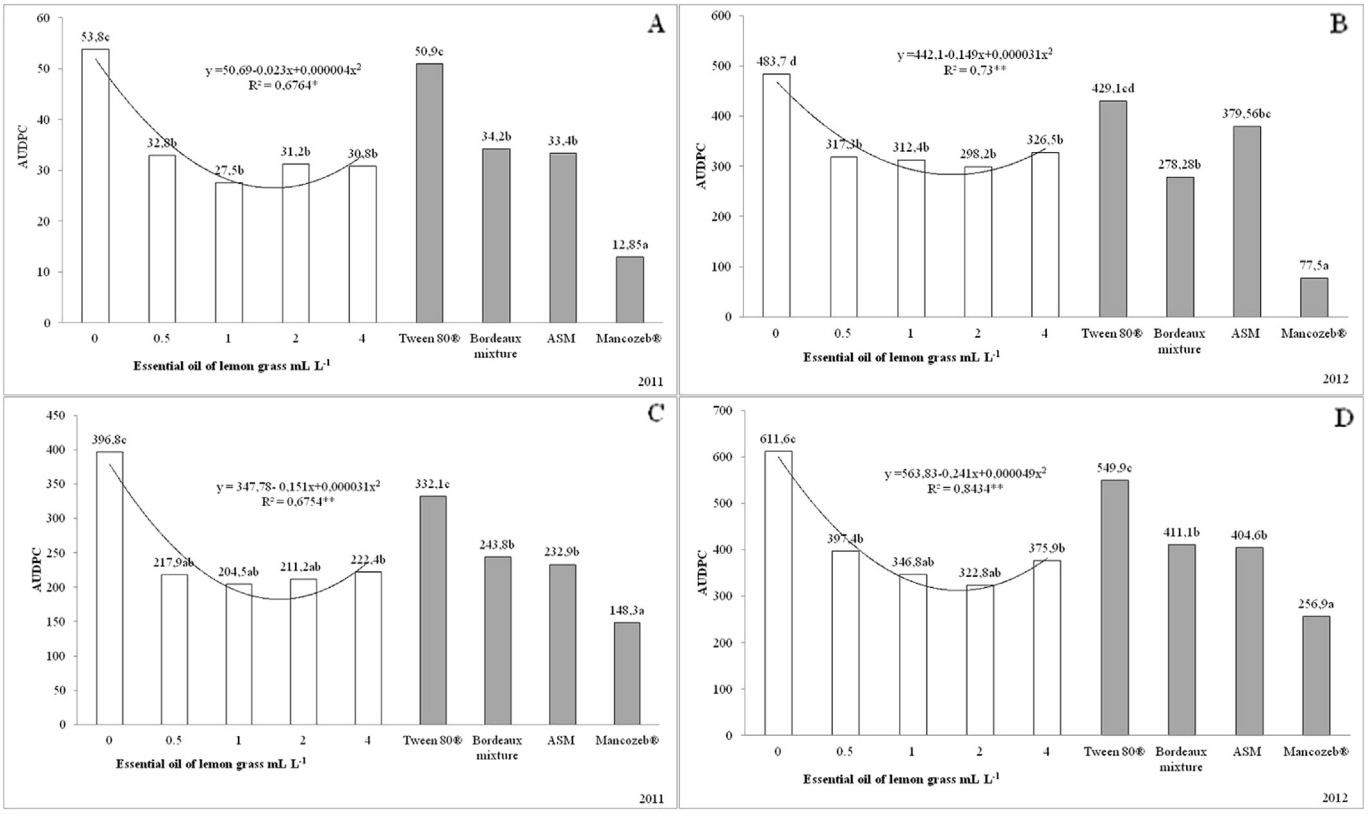


Fig. 1. Effects of doses of essential oil of lemon grass on the AUDPC, leaf spot (A and B) and mildew (C and D) on grapevine cv. Isabel in two consecutive years. **Significant ($p = 1\%$). Means followed by the same letter do not differ (Tukey test, $p = 5\%$).

conventional production to organic or integrated production (Detoni et al., 2005).

Accordingly, it is necessary to use alternative methods to control diseases that occur in organic agriculture. Among these alternative methods are the induction of resistance in plants (Bonaldo et al., 2005) and the use of natural bioactive substances with antimicrobial activity (Camili et al., 2007). In this context, the use of essential oils extracted from medicinal plants that contain secondary compounds can have both fungitoxic actions (antimicrobial action) and facilitate resistance by activating defense mechanisms in plants (indirect antimicrobial action). For these reasons, the use of these essential oils may represent a promising alternative in organic agriculture (Franzener et al., 2003).

Lemon grass (*Cymbopogon citratus* (DC) Stapf), a member of the family Poaceae, is a medicinal plant that contains compounds that can potentially control pathogen-caused plant disease and/or induce plant resistance to pathogens. This aromatic plant is cultivated for the commercial production of an essential oil that is widely used as a scenting agent in perfumery and cosmetics, in the preparation of colognes, soaps and deodorants and in the pharmaceutical industry (Costa et al., 2005). Its typical major constituents are citral monoterpenes (an isomeric mixture of neral and geranial) and myrcene (Guimarães et al., 2011). According to Souza et al. (1991), the citral monoterpenes show antimicrobial and antifungal activity, and these properties are currently attracting attention in agronomy.

Pereira et al. (2012) evaluated the activity of essential oils of *Cinnamomum zeylanicum* (cinnamon), *C. citratus* (lemon grass), *Syzygium aromaticum* (clove), *Melaleuca alternifolia* (tea tree), *Thymus vulgaris* (thyme), *Azadirachta indica* (neem), *Corymbia citriodora* (eucalyptus) and *Cymbopogon nardus* (citronella grass) to control rust (*Hemileia vastratrix* Berk. & Br) in coffee seedlings in a greenhouse. The essential oil of lemon grass was found to reduce the severity of the disease by 67% and 70% in Catuaí IAC62 and Catuaí 2SL, respectively, demonstrating the effectiveness of the essential oil in controlling this disease.

The effect of the essential oil of *C. citratus* on induced resistance has been verified by Balbi-Peña et al. (2007), who used the oil to treat tomato plants inoculated with *Alternaria solani*. In that study, increased peroxidase activity at 12 and 48 h after inoculation

Table 1
Total soluble solids (TSS), pH, and the weight of bunches of cv. Isabel grapes subjected to different doses of essential oil from lemon grass in crops 2011. Marialva-PR.[a]

Treatments	Total soluble solids (TSS)	pH	Weight of bunches (g)
Control	15.03bc[1]	3.1[ns;2]	60.94 [ns;3]
Essential oil lemon grass 0.5 mL L^{-1}	16.00ab	3.2	63.79
Essential oil lemon grass 1.0 mL L^{-1}	16.55a	3.1	60.98
Essential oil lemon grass 2.0 mL L^{-1}	13.63cd	3.0	52.72
Essential oil lemon grass 4.0 mL L^{-1}	15.73ab	3.1	52.37
Tween 80%	13.95cd	3.1	58.93
Bordeaux mixture (1:1:1)	14.40cd	2.9	57.52
S-methyl-Acibenzolar	14.27cd	3.1	57.33
Mancozeb	15.9 ab	4.9	64.98
CV%	4.36	40.2	14.03
Fc	13.36	1.15	0.22

Means followed by the same letter do not differ by Tukey test at 5% probability not significant for the average test. Tukey and polynomial regression at the level of 5%. [1]Pr > $Fc_{regression}$ = 0.000/Pr > t = 0.0134/y = 15.43−0.00007x/R^2 = 0.011. [2]Pr > $Fc_{regression}$ = 1.737/CV% = 2.83. [3]Pr > $Fc_{regression}$ = 0.1681/CV% = 14.96.
[a] Marialva, city located in the state of Paraná (PR), Brazil.

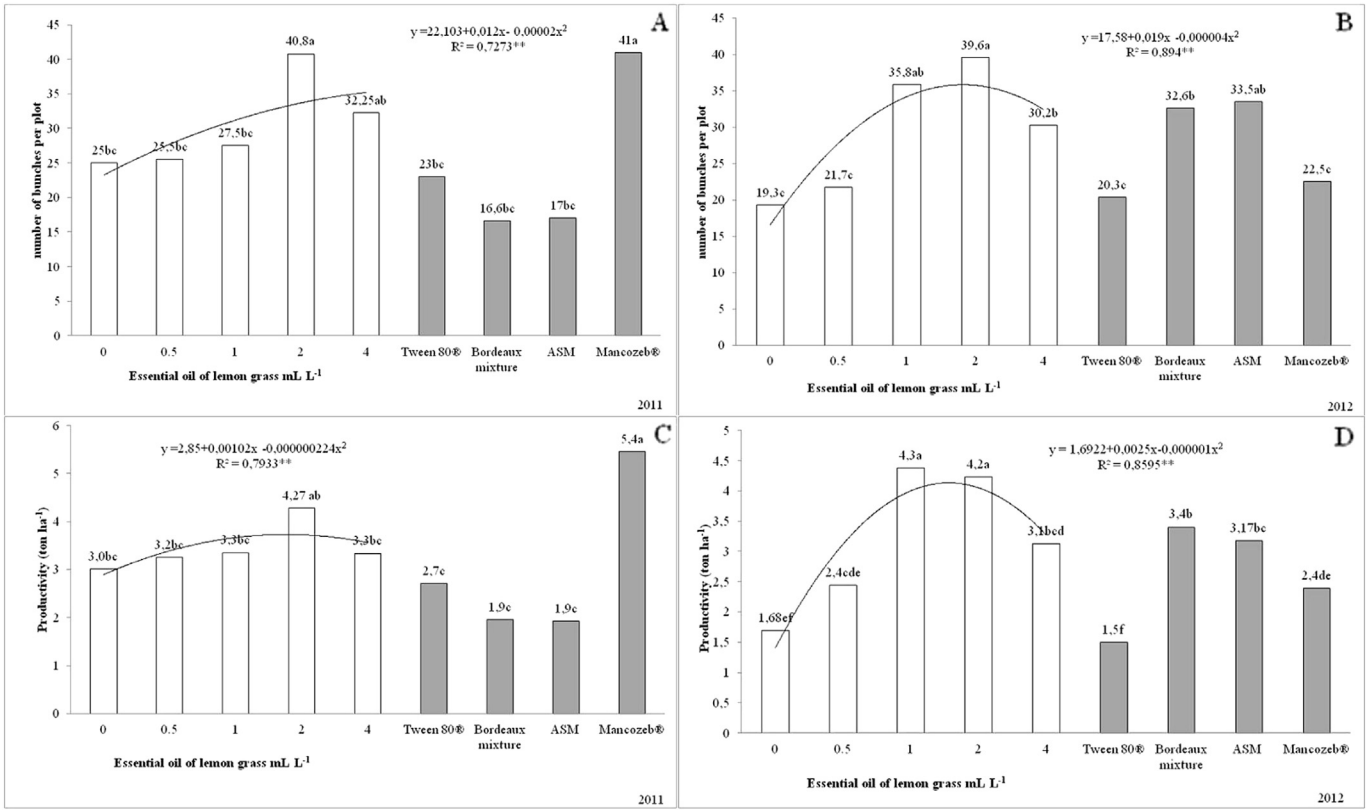


Fig. 2. Number of clusters (A and B) and productivity (C and D) of cv. Isabel exposed to different concentrations of essential oil of lemon grass for 2011 and 2012 harvests. **Significant ($p = 1\%$). Means followed by the same letter do not differ (Tukey test, $p = 1\%$).

compared with plants treated with water indicated the potential ability of *C. citratus* essential oil to induce disease resistance in plants.

In view of these findings, the aim of this study was to evaluate the potential ability of the essential oil of lemon grass to control isariopsis leaf spot (*P. vitis*) and downy mildew (*P. viticola*) and to influence productivity and enzyme activity in grapevine cv. Isabel (*Vitis labrusca* L.).

2. Material and methods

The experiment was conducted in a commercial vineyard with cv. Isabel under the conventional culture system during two consecutive cycles in 2011 (August/December) and 2012 (February/July). The vines, in their eighth year of production, were grafted on rootstock '420-A', spaced at 2.0 × 2.5 m and trained on a pergola system. The vineyard was located in the municipality of Marialva, Paraná, Brazil (23° 27′ 49.86″S, 51° 47′ 18.74″W) at an altitude of 614 m. The climate is Cfa (humid sub-tropical) according to the Köppen classification (SIMEPAR, 2012). The soil at the site selected for the experiment was classified as a typic eutrophic and eutrophic red clay soil of loamy soil, and the terrain ranged from flat to rolling (EMBRAPA, 2006).

The essential oil of lemon grass was obtained from dried leaves, previously kept at room temperature, by steam drag distillation at 120 °C for two hours. The molecules of essential oils evaporate along with water vapor through the distiller, are immediately cooled and condense along with the water. The water layer and the essential oil differ in density and are separated with a dropping funnel (Worwood, 1995). The lemon grass leaves were obtained from the orchard at the Experimental Farm of Iguatemi-State University of Maringá (UEM). Doses of 0, 0.5, 1.0, 2.0 and 4.0 mL L^{-1} essential oil were applied to the grape plants. To solubilize the oil, Tween® 80% was used in a 1:1 proportion of essential oil: Tween® 80%. To compare the plants in these five treatments with plants subjected only to a standard treatment, standard treatments with Tween® 80%, Bordeaux mixture, Acibenzolar-S-methyl and mancozeb were applied to different plants. The standard treatments were applied as follows. Bordeaux mixture was used in a 1:1:100 proportion of copper sulfate: lime: water. Acibenzolar-S-methyl (S-methyl-Acibenzolar) was used at 50 mg L^{-1}, and mancozeb (Dithane®) was applied at 3 kg of formulated product per hectare.

The treatments were applied weekly with a hand sprayer during the cooler hours of the day. Spraying was performed until liquid was observed to drip from the plants. The treatments were applied from the beginning of sprouting on 30/08/2011 and 14/02/2012, a total of 10 applications. The experimental design was a randomized block with nine treatments and five replicates, and one plant was used to represent a plot.

The plants were observed to detect the onset of symptoms. The natural occurrence and severity of the symptoms, the presence of isariopsis leaf spot (after the fourth application of the treatment in both crop cycles) and the presence of downy mildew (after the seventh application of the treatment in both crop cycles) were evaluated based on observations of three leaves from the median of four previously identified branches per plant. Diagrammatic scales were used to evaluate the presence of the diseases. A diagrammatic scale of one to six, corresponding to a range of lesions of 1.6% (i.e., lesions present on 1.6% of the leaves examined) to 40.20% (Lenz et al., 2009) was used to evaluate the presence of isariopsis leaf spot; a diagrammatic scale of one to twelve, corresponding to a range of lesions of 0–100%, was used to evaluate the presence of downy mildew (Azevedo, 1997). Disease severity was determined

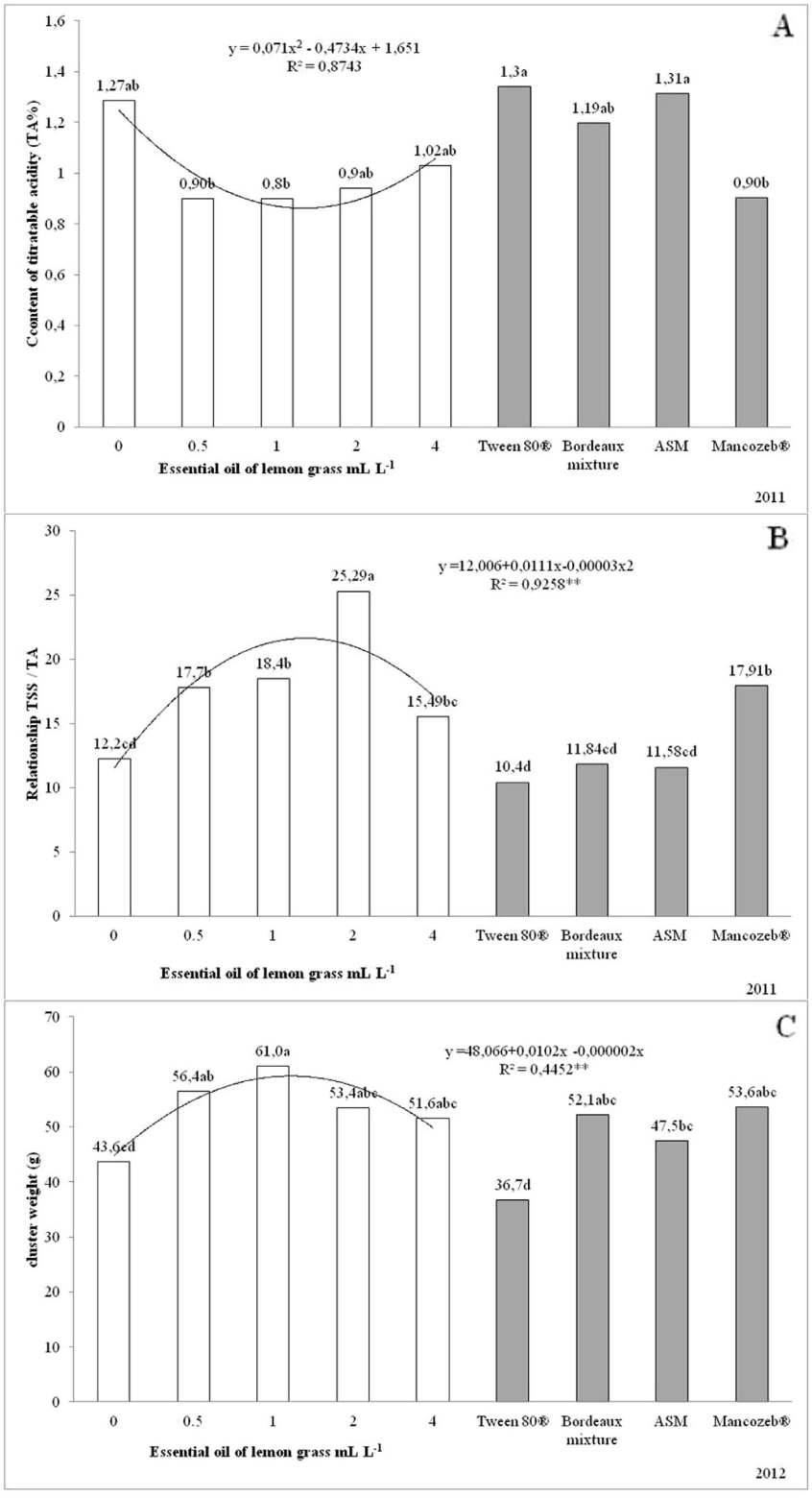


Fig. 3. Titratable acidity expressed as % of tartaric acid (A), maturation index (SST/AT) (B) and weight of clusters (g) (C) of grapevines cv. Isabel exposed to different concentrations of essential oil of lemon grass. **Significant ($p = 1\%$). Means followed by the same letter do not differ (Tukey test, $p = 1\%$).

by the area under the disease progress curve (AUDPC). The AUDPC was calculated for the time period between the day on which the disease was initiated and the twenty-fifth day after the initiation of the disease. The AUDPC was calculated as follows according to the method described by Campbell and Madden (1990). To plot the disease progress curve, a total of five evaluations were performed at intervals of seven days. The following equation (Campbell and Madden, 1990) was used to calculate the AUDPC:

Table 2
Titratable acidity (% of TA), total soluble solids (SST), pH of cv. Isabel grapes exposed to different concentrations of essential oil from lemon grass in crops 2012. Marialva-PR.[a]

Treatments	Titratable acidity (% of TA)	Total soluble solids (SST)	SST/ATT	pH
Control	0.73[ns;1]	12.43ab[2]	18.81ab[3]	3.1ab[4]
Essential oil lemon grass 0.5 mL L^{-1}	0.64	13.48ab	21.53ab	3.0ab
Essential oil lemon grass 1.0 mL L^{-1}	0.74	12.98ab	17.78b	3.0ab
Essential oil lemon grass 2.0 mL L^{-1}	0.65	11.46bab	17.97b	3.0ab
Essential oil lemon grass 4.0 mL L^{-1}	0.71	12.37ab	17.66b	3.11a
Tween 80%	0.69	12.72ab	18.62ab	3.0ab
Bordeaux mixture (1:1:1)	0.60	13.3ab	21.9ab	3.12a
S-methyl-Acibenzolar	0.60	10.96b	18.35ab	3.0ab
Mancozeb	0.58	14.86a	25.99a	2.92b
CV%	18.02	11.30	18.77	2.5
Fc	1.15	3.16	2.8	2.89

Means followed by the same letter do not differ by Tukey test at 5% probability not significant for the average test. Tukey and the level of polynomial regression 5%. [1]Pr > $Fc_{regression}$ = 0.716/Fc = 0.528/Cv% = 20.58; [2]Pr > $Fc_{regression}$ = 0.2632/Fc = 1.451/CV% = 11.2; [3]Pr > $Fc_{regression}$ = 0.4758/Fc = 0.922/CV% = 20.06; [4]Pr > $Fc_{regression}$ = 0.3591/Fc = 1.174/CV% = 2.73.

[a] Marialva, city located in the state of Paraná (PR), Brazil.

$$\text{AUDPC} = \sum_{i}^{n-1}\left[\left(\frac{y_i + y_{i+1}}{2}\right)(t_{i+1} - t_i)\right]$$

In this equation, n is the total number of assessments, t_i is the time at which evaluation i was performed, and y_i is the severity of the disease at time t_i. The percent reduction of disease was determined for each treatment in comparison to the control. The control was considered to represent a severity of 100%.

Additionally, assessments of productivity and grape quality were performed in two consecutive years. The variables analyzed to calculate the value of productivity were the number of clusters per plot and the average weight of the clusters. Productivity (tons ha^{-1}) was estimated from these data. To determine the physical and chemical characteristics of the fruits, three fruits per plot were collected. To represent each plot, the three fruits from that plot were placed in a centrifugal home juice extractor and crushed. A Brix degree scale, based on specific gravity at 16 °C/16 °C, was used to measure the content of total soluble solids (TSS). The TSS content was determined using a portable refractometer and expressed in °Brix. The titratable acidity (TA) was obtained by titration of the juice with a 0.1 N standardized solution of NaOH. A pH of 8.2 was used as the endpoint of the titration. The result was expressed as a percentage of tartaric acid (Instituto Adolfo Lutz, 1985). The pH was measured with a TEC-11 pH meter.

For biochemical analysis, the collection of plant material (leaves) was performed in five stages. For the 2011 harvest, leaves were collected 24 h (first collection) and 48 h (second collection) after the second application, 24 h before the third application (third collection) and 24 h (fourth collection) and 48 h (fifth collection) after the third application. For the 2012 harvest, leaves were collected at 24 h (first collection) and 48 h (second collection) hours after the fourth application, 24 h before the fifth application (third collection), 24 h (fourth collection) and 48 h (fifth collection) after the fifth application. Samples were collected from the plants that received doses of essential oil (0, 0.5, 1.0, 2.0 and 4.0 mL L^{-1}) and from the plants treated with Tween® 80% and Acibenzolar-S-methyl.

Enzyme extracts were obtained by macerating 1 g of vine leaves in liquid nitrogen and then mechanically homogenizing the macerated leaves in 4 mL of 50 mM potassium phosphate buffer (pH 7.0) containing 0.1 mM EDTA and 1% (p/p) polyvinylpyrrolidone (PVP) in a mortar. The homogenate was centrifuged at 15,000 *g* for 30 min at 4 °C. The supernatant, representing enzyme extract, was stored at −20 °C prior to analysis.

The protein content was determined according to Bradford (1976). A total of 2.5 mL of Bradford reagent was added, with stirring, to 50 μL of the supernatant. After 5 min, the absorbance at 595 nm was read with a spectrophotometer. The protein concentration in mg per mL of sample (mg protein mL^{-1}) was determined using standard curve concentrations of bovine serum albumin (BSA) from 0 to 0.5 mg mL^{-1}. The standard curve for bovine serum albumin was obtained by the methods specified by Bradford (1976).

Chitinase (EC 3.2.1.14) was assessed based on the release of soluble fragments of liquid CM-chitin-RBV (carboxymethyl-chitin-Remazol Brilliant Violet) from carboxymethylated chitin labeled with Remazol brilliant violet (Loewe Biochemica GmbH) (Wirth and Wolf, 1990). For this determination, a total of 600 μL of extraction buffer (100 mM sodium acetate, pH 5.2) was mixed with 200 μL of enzyme extract and 200 μL of CM-chitin-RBV (2 mg L^{-1}). After 20 min of incubation at 40 °C, the reaction was quenched with 200 μL 1 M hydrochloric acid (HCl) followed by cooling on ice and centrifugation at 10,000 *g* for 5 min. The absorbance of the supernatant was determined at 550 nm. The results are expressed in absorbance units min^{-1} mg^{-1} protein.

Catalase (EC 1.11.1.6) activity was quantified with the method of Góth (1991), modified by Tomanková et al. (2006), using the stable complex formed by ammonium molybdate and hydrogen peroxide (A405). The enzyme extract (0.2 mL) was incubated in 1 mL reaction mixture containing 60 mM hydrogen peroxide in 60 mM pH 7.4 potassium phosphate buffer at 38 °C for 4 min. A total of 1 mL of 32.4 mM ammonium molybdate was added after 4 min of incubation to end the consumption of hydrogen peroxide by the enzyme present in the extract. A blank was prepared for each sample by adding ammonium molybdate to the reaction mixture and omitting the incubation period. The yellow complex of molybdate and hydrogen peroxide was measured at 405 nm. The difference between the absorbance of the blank and the absorbance of the incubated sample indicated the amount of hydrogen peroxide used by the enzyme. The concentration of hydrogen peroxide was determined using a value of $\epsilon = 0.0655$ mM^{-1} cm^{-1} for the extinction coefficient.

A qualitative analysis of the essential oils was performed with gas chromatography coupled to mass spectrometry (GC–MS). The analysis used a Thermo Electron FOCUS GC chromatograph equipped with a DB-5ms fused silica capillary column (30 m × 0.25 mm, 0.25 mm in stationary phase) and a Thermo Electron DSQ II mass spectrometer. The operating conditions were as follows: injector temperature 200 °C, flow rate of carrier gas (helium) 1.0 mL/min, injection volume 0.5 μL (dichloromethane solution) to flow division ("split") in a ratio of 1:40. The oven temperature program was 40 °C for 6 min, then increased to 180 °C at a rate of 3 °C/min and then from 180 °C to 220 °C at a rate of 10 °C/min, finally remaining at 220 °C for 20 min. Mass spectra were obtained via electron impact ionization at 70 eV in a range of 40–650 *m/z* with an interface temperature of 250 °C and a 200 °C ion source. The identification of compounds was based on the Kovats Retention Index (KI), determined with respect to the homologous series of n-alkanes analyzed under the same

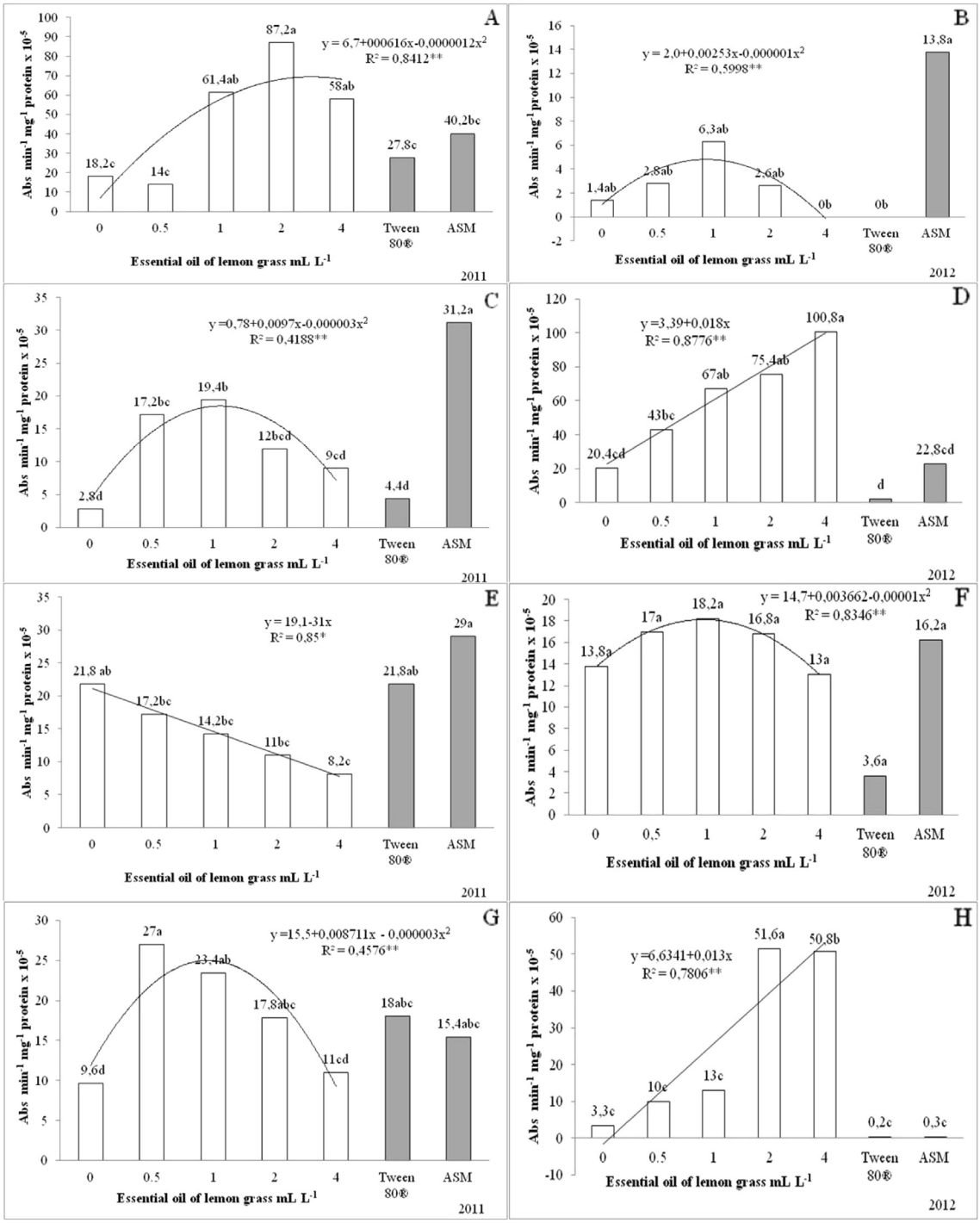


Fig. 4. Dynamics of chitinase in leaves of grapevine cv. Isabel exposed to various concentrations of essential oil of lemon grass. Harvest in 2011: A = 24 h after the second spraying (first collection), C = 48 h after the second spraying (second collection), E = 24 h after the third spraying (fourth collection) and G = 48 h after the third spraying (fifth collection). Harvest in 2012: B = 24 h after the third spraying (first collection), D = 48 h after the third spraying (second collection); F = 24 h after the fourth spraying (fourth collection) and H = 48 h after the fourth spraying (fifth collection). ** Significant (p = 1%). Means followed by the same letter do not differ (Tukey test, p = 5%).

experimental conditions, a search of the MS library (NIST) and comparisons with data in the literature (Guimarães et al., 2011). Quantitative data were obtained from the percentages of the areas of the chromatogram after standardization.

The results were evaluated with an analysis of variance. Significant results were analyzed with a Tukey test to compare means and with a polynomial regression. A 5% significance level was used for all tests. All calculations were performed with the SISVAR statistical program (Ferreira, 2011).

3. Results and discussion

The AUDPC analysis showed that the essential oil of lemon grass affected the severity of leaf spot in both of the two consecutive cycles examined (Fig. 1A and B). The application of 0.5, 1.0, 2.0 and 4.0 mL L^{-1} of essential oil produced decreases in the AUDPC of approximately 40%, 48%, 42% and 42%, respectively, an apparent quadratic effect. The effects of the Bordeaux mixture and Acibenzolar-S-methyl treatments did not differ and were not as

Table 3
Dynamics of chitinase in leaves of cv. Isabel grapevines exposed to different concentrations of essential oil from lemon grass 24 h before pulverization in the third and fourth seasons, respectively. Marialva-PR.[a]

Treatments	Crops 2011	Crops 2012
	Chitinase	Chitinase
Control	1.2b[1]	0.1c[2]
Essential oil lemon grass 0.5 mL L^{-1}	14.4a	19.8a
Essential oil lemon grass 1.0 mL L^{-1}	2.8b	1.2c
Essential oil lemon grass 2.0 mL L^{-1}	1.4b	7.0b
Essential oil lemon grass 4.0 mL L^{-1}	3.2b	0.4c
Tween 80%	2.2b	0.7c
S-methyl-Acibenzolar	6.6b	1.4c
CV (%)	76.34	57.42
Fc	9.368	41.25

Means followed by the same letter do not differ by the Tukey test at 5% level of probability. Chitinase 550 nm $min^{-1}mg^{-1}$protein $\times 10^{-5}$; [1]Pr > Fc = 0.001/Pr > t_{linear} = 0.091/Pr > $t_{quad.}$ = 0.668; [2]Pr > Fc = 0.0000/Pr > $t_{quad.}$ = 0.005. $y = 0.000059 + 0.00000003x + 0.000000x^2/R^2 = 0.093$.

[a] Marialva, city located in the state of Paraná (PR), Brazil.

strong as the effects of the chemical treatment, which decreased the AUDPC by 76% for the 2011 harvest (Fig. 1A). In the second cycle, in 2012, the doses of 0.5, 1.0, 2.0 and 4.0 mL L^{-1} essential oil of lemon grass did not show significantly different effects on the AUDPC. These doses resulted in decreases of approximately 34%, 35%, 38% and 32%, respectively (Fig. 1B).

The essential oil of lemon grass also showed a quadratic effect on the AUDPC for mildew for the 2011 harvest. Doses of 0.5, 1.0 and 2.0 mL L^{-1} of the essential oil of lemon grass decreased the AUDPC by 45%, 48% and 46%, respectively. These values did not differ from the effects of treatment with Bordeaux mixture, Acibenzolar-S-methyl or mancozeb (Fig. 1C). Similar results were obtained in the second year. Doses of 1.0 and 2.0 mL L^{-1} essential oil of lemon grass reduced the AUDPC by approximately 43% and 48%, respectively, compared with the control. These values did not differ significantly from the 58% decrease in the AUDPC produced by the mancozeb treatment (Fig. 1D). The results of this study also showed that the 80% Tween used as an adjuvant did not influence the control of isariopsis leaf spot or downy mildew.

Few research reports have addressed the use of medicinal plants in the control of grapevine diseases based on experiments conducted in the field. However, results similar to those of the present study were obtained by Pereira et al. (2012), who evaluated the effectiveness of essential oils of *C. zeylanicum* (cinnamon), *C. citratus* (lemon grass), *S. aromaticum* (clove), *M. alternifolia* (tea tree), *T. vulgaris* (thyme), *A. indica* (neem), *C. citriodora* (eucalyptus) and *C. nardus* (citronella grass) in the control rust in coffee seedlings in a greenhouse. The results of this previous study showed that a dose of 1.0 mL L^{-1} of essential oil reduced the severity of the disease by 67% and 70% in Catuaí IAC62 and Catuai 2SL, respectively. The essential oils of lemon grass, tea tree, thyme, citronella grass, eucalyptus and neem at a dose of 1.0 mL L^{-1} have been found to reduce the AUDPC of brown eyespot (caused by *Cercospora coffeicola* Berk & Cooke) by 27.68%, 20.36%, 18.39%, 18.03%, 16.95% and 15.43%, respectively, in coffee plants cv. Catuaí 2SL maintained in a greenhouse (Pereira et al., 2011).

The results of the present study showed that the severity of disease, as measured by the AUDPC, was greater for the off-season harvest (2012). It is possible that the results for 2011, the previous year of the cycle, were influenced by the prevailing climatic conditions.

The major portion of the productive cycle of the grapevine occurred in the rainy season of 2012 (510 mm total rainfall accompanied by a relative humidity of approximately 70% – SIMEPAR, 2012), favoring an increase in the severity of leaf spot and mildew.

Accordingly, this study shows that the essential oil of lemon grass is effective in controlling grapevine diseases even under conditions that favor disease development. In terms of the physical characteristics of the crop, the sample taken from the first harvest at 123 days after pruning showed that the doses of the essential oil had no significant effect on the mass of the clusters evaluated (Table 1). However, the number of clusters per plant and the yield were affected by the treatment with the essential oil. The maximum values of these two variables occurred at estimated doses of the essential oil of lemon grass of 2.98 and 2.57 mL L^{-1}, respectively (Figs. 2A and C). A dose of 2.0 mL L^{-1} of essential oil of lemon grass increased the cluster number and yield by 63% and 48%, respectively, compared with the control. These values did not differ significantly from those produced by the chemical treatment.

In the second harvest, performed 127 days after pruning, a quadratic effect of the dose of the essential oil of lemon grass was found on the number of clusters, the cluster weight and the yield. The estimated doses corresponding to the maximum values of these three variables were 2.37, 1.25 and 2.55 mL L^{-1}, respectively (Figs. 2B, D and 3C). Of the doses tested, 2.0 mL L^{-1} essential oil of lemon grass increased the number of clusters per plant by 92% and the yield by 150%. These values differed significantly from those produced by Bordeaux mixture and by mancozeb (Figs. 2B and D).

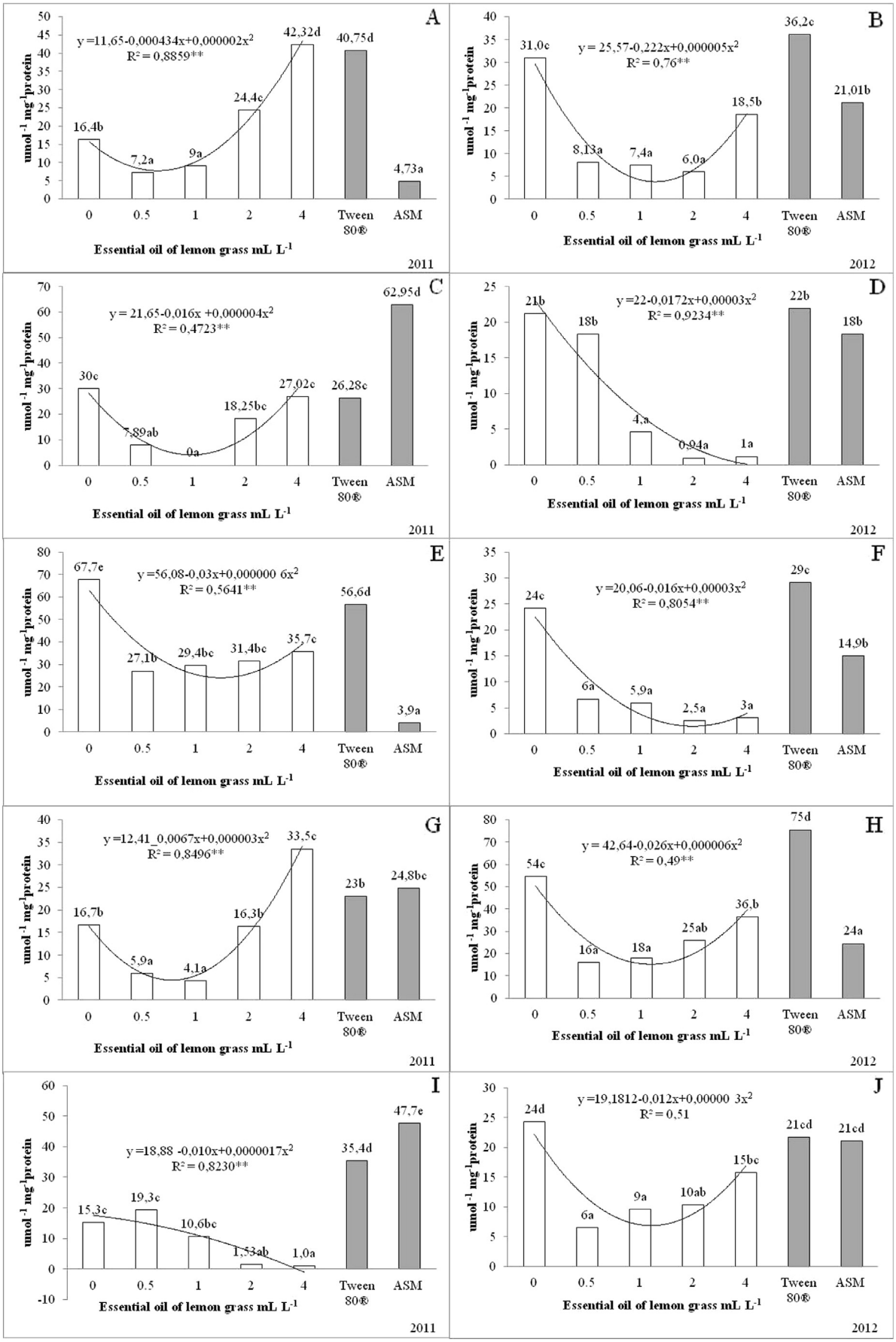
In terms of the chemical characteristics of the grapes, the doses of essential oil of lemon grass had no significant effect on pH in 2011 (Table 1). However, the dose of essential oil had a quadratic effect on TA, TSS and TSS/TA (Fig. 3 and Table 1). A dose of 0.5 mL L^{-1} of essential oil of lemon grass caused the acidity to decrease by 37% compared with the control. This value did not differ significantly from that produced by the mancozeb treatment (Fig. 3A). The maximum values of the soluble solids and maturation index were observed at estimated doses of 2.26 and 1.85 mL L^{-1}, respectively. A dose of 2.0 mL L^{-1} caused an increase of 100% in the maturation index relative to that found for the control treatment. This 100% increase differed significantly from the 46% increase in the maturation index produced by mancozeb, suggesting an effect prior to maturation (Fig. 3B).

For the 2012 harvest, doses of the essential oil of lemon grass did not affect the chemical characteristics of the grapes. The effect of these doses did not differ significantly from those of the standard treatments (Table 2). The treatment with 80% Tween did not influence the physicochemical properties of the grapes.

According to Rizzon and Link (2006), the maturation index represents the balance between the sweet and sour tastes of grape juice. Specifically, higher values of the maturation index indicate that the juice tastes sweeter. This balance reflects the relationship between TSS and TA. Accordingly, it appears that, although the TSS obtained in the second year of production was not high, the maturation index was within the optimal range for obtaining quality juices. Rizzon et al. (1998) pointed out the maturation index as the most important chemical characteristic to consider at the time of the selection of grapes for processing. In the present study, the grapes had TA levels during the growing off-season that were lower than those obtained in the 2011 harvest. Colombo et al. (2011) suggest that the development of acidity in the grape is due primarily to the degradation of malic acid, which is favored by higher air temperature and the stage of maturation of the grapes.

In this context, the plants treated with the essential oil of lemon grass showed improvement in two consecutive years in the characteristics of the grape related to the quality of the fruit for use in wine production.

The essential oil altered the metabolism-based disease resistance in grapevine, in part by causing an increase in chitinase activity. For 2011, the dose of the essential oil of lemon grass showed quadratic effects on the activity of chitinase for all evaluation


A
y =11,65-0,000434x+0,000002x²
R² = 0,8859**
16,4b
7,2a
9a
24,4c
42,32d
40,75d
4,73a
umol -1 mg-1protein
Essential oil of lemon grass mL L-1
Tween 80®
ASM
2011
B
y = 25,57-0,222x+0,000005x²
R² = 0,76**
31,0c
8,13a
7,4a
6,0a
18,5b
36,2c
21,01b
2012
C
y = 21,65-0,016x +0,000004x²
R² = 0,4723**
30c
7,89ab
0a
18,25bc
27,02c
26,28c
62,95d
2011
D
y =22-0,0172x+0,00003x²
R² = 0,9234**
21b
18b
4,a
0,94a
1a
22b
18b
2012
E
y =56,08-0,03x+0,000000 6x²
R² = 0,5641**
67,7e
27,1b
29,4bc
31,4bc
35,7c
56,6d
3,9a
2011
F
y =20,06-0,016x+0,00003x²
R² = 0,8054**
24c
6a
5,9a
2,5a
3a
29c
14,9b
2012
G
y =12,41_0,0067x+0,000003x²
R² = 0,8496**
16,7b
5,9a
4,1a
16,3b
33,5c
23b
24,8bc
2011
H
y = 42,64-0,026x+0,000006x²
R² = 0,49**
54c
16a
18a
25ab
36,b
75d
24a
2012
I
y =18,88 -0,010x+0,0000017x²
R² = 0,8230**
15,3c
19,3c
10,6bc
1,53ab
1,0a
35,4d
47,7e
2011
J
y =19,1812-0,012x+0,00000 3x²
R² = 0,51
24d
6a
9a
10ab
15bc
21cd
21cd
2012

periods except 24 h after the third spray, which showed a linear effect of the essential oil on chitinase activity (Fig. 4, C, E and G). The maximum values of chitinase activity were obtained for estimated doses of 2.57, 1.62 and 1.45 mL L^{-1} for the periods 24 and 48 h after the second spray and 48 h after the third spraying, respectively. The greatest percentage (300% and 181%) increase in chitinase activity was observed at a dose of 2.0 mL L^{-1} 24 h after the second application and at a dose of 0.5 mL L^{-1} 48 h after the third spraying, respectively. These values differed significantly from those produced by Acibenzolar-S-methyl, which increased the chitinase activity 219% and 60% compared with the control. No significant regression relationship between chitinase activity and the dose of essential oil was found 24 h before the third spraying, but the dose of 0.5 mL L^{-1} showed higher chitinase activity that differed significantly from the control value and from the value for the Acibenzolar-S-methyl treatment (Table 3).

In the second harvest, the dose of essential oil of lemon grass had a quadratic effect on chitinase activity 24 h after spraying (Fig. 4B and F). A dose of 2.0 mL L^{-1} 24 h after the third spraying produced chitinase activity that was four times greater than the control value. This increase was less than that found for the Acibenzolar-S-methyl treatment, which produced a level of chitinase activity ten times greater than that found for the control (Fig. 4B). The maximum chitinase activity 24 h after the fourth spraying resulted from an estimated dose of 1.83 mL L^{-1} (Fig. 4F). A dose of 1.0 mL L^{-1} increased chitinase activity by approximately 37% compared with the control. However, no significant difference was found among the treatments. No significant regression relationship between chitinase activity and the dose of essential oil was found 24 h before the fifth spraying. However, a dose of 0.5 mL L^{-1} resulted in chitinase activity that was significantly higher than the values for the control and for the Acibenzolar-S-methyl treatment (Table 3). For samples taken 48 h after spraying, the dose of the essential oil had a positive linear effect on chitinase activity (Fig. 4D and H). Forty-eight hours after the second and third sprayings, the highest dose of the essential oil produced an increase in chitinase activity that was five to fifteen times greater, respectively, than the corresponding values found for the control treatment. The increases resulting from this dose of the essential oil differed significantly from the corresponding increases resulting from the Acibenzolar-S-methyl treatment. The dose of 0.5 mL L^{-1} produced high chitinase activity even after six days of treatment application in two consecutive cycles, showing that the expression of the resistance induced by the plant can be observed for a longer time, as previously found by Kuc (1995).

The antimicrobial activity of chitinase is based on an ability to hydrolyze chitin polymers, weakening the cell wall and making the cells osmotically sensitive (Selitrennikoff, 2001). It is possible that the effects of this mechanism are responsible for the reduction in the severity of isariopsis leaf spot on the grape leaves. However, one of the primary characteristics of the causal agent of downy mildew, an obligate parasite belonging to the class Oomycetes, is an absence of chitin in the cell wall (Massola Júnior and Krugner, 2011). In addition to the antimicrobial effects of chitinase, according to Yoshikawa et al. (1993), this enzyme acts indirectly by releasing fragments of the material components of plant cell walls (oligogalacturonides). These fragments serve to elicit defense responses in the host plant. Most likely, the treatments with the essential oil of lemon grass in this study stimulated other mechanisms, e.g., the metabolic formation of phytoalexins, to reduce the severity of downy mildew on the grapevines. Poinssot et al. (2012) suggest that secondary metabolites such as resveratrol, derivatives and flavonoids contribute to cell wall strengthening in grapevines in response to infection or treatment with agents that elicit defensive responses.

The doses of 0.5 and 1.0 mL L^{-1} of essential oil of lemon grass produced the lowest catalase activity, approximately 60% less than that observed for the control (Fig. 5A, C and G). These values of catalase activity did not differ from those resulting from the Acibenzolar-S-methyl treatment 24 h after the second spraying (Fig. 5A). Forty-eight hours after the second spraying, a dose of 1.0 mL L^{-1} of the essential oil of lemon grass produced no catalase activity. This result differed significantly from that found for the Acibenzolar-S-methyl treatment, which produced higher catalase activity (Fig. 5C). In the third collection, performed 24 h before the third spraying, the effect of the essential oil was less than the effect of the Acibenzolar-S-methyl treatment, which decreased catalase activity by approximately 94% (Fig. 5D). Twenty-four hours after the third spraying, the lower doses of the essential oil also produced lower catalase activity, a decrease of approximately 70% relative to the control. This value differed significantly from that produced by the Acibenzolar-S-methyl treatment (Fig. 5G). The doses of 2.0 and 4.0 μL L^{-1} essential oil decreased catalase activity by approximately 90% and 93%, respectively, relative to the control. These decreases differed significantly from those produced by the Acibenzolar-S-methyl treatment (Fig. 5I).

For the second harvest, the doses of essential oil showed a quadratic effect on catalase activity at all times of evaluation (Fig. 5B, D, F, H and J). The doses of 0.5, 1.0 and 2.0 mL L^{-1} showed less activity, producing decreases in catalase activity of 73%, 76% and 80%, respectively, compared with the control. These values differed significantly from those produced by the Acibenzolar-S-methyl treatment, which decreased catalase activity by 32% 24 h after the third spraying (Fig. 5A). Forty-eight hours after spraying, doses of 1.0, 2.0 and 4.0 mL L^{-1} of essential oil caused decreases in catalase activity of 80%, 95% and 95%, respectively, compared with the control. These values differed significantly from the corresponding values for the other treatments (Fig. 5D). In the third sample, taken 24 h before spraying, treatment with essential oil decreased catalase activity by 89%–75% compared with the control. In contrast, the Acibenzolar-S-methyl treatment produced a smaller decrease, 41% (Fig. 5F). Essential oil doses of 0.5, 1.0 and 2.0 mL L^{-1} caused decreases in catalase activity of approximately 70%, 66% and 53%, respectively, compared with the control. These values did not differ significantly from the values obtained with the Acibenzolar-S-methyl treatment 24 h after spraying (Fig. 5H). Additionally, the values of catalase activity obtained for the essential oil doses of 0.5 and 1.0 mL L^{-1} 48 h after spraying did not differ significantly from the corresponding values obtained with the Acibenzolar-S-methyl treatment.

The plants treated with Acibenzolar S-methyl showed a significant increase in catalase activity 48 h after the second spraying and 48 h after the third spraying (Fig. 5A and I). Similarly, the plants treated with various doses of essential oil showed increased catalase activity 24 h before the third spraying (Fig. 5E). During the second cycle, the plants treated with Tween 80® showed increased catalase activity for all periods (Fig. 5B, D, F, H and J). This result appears to represent a response to the extremely high levels of hydrogen peroxide that were most likely present.

Fig. 5. Dynamics of catalase in leaves of grapevine cv. Isabel exposed to various concentrations of essential oil of lemon grass. Harvest in 2011: A = 24 h after the second spraying (first collection), C = 48 h after the second spraying (second collection), E = 24 h before the third spraying (third collection) and G = 24 h after the third spraying (fourth collection), I = 48 h after the third spraying (fifth collection). Harvest in 2012: B = 24 h after the fourth spraying (first collection), D = 48 h after the fourth spraying (second collection), F = 24 h before the fifth spraying (third collection), H = 24 h after the fifth spraying (fourth collection) and J = 48 h after the fifth spraying (fifth collection). ** Significant (p = 1%). Means followed by the same letter do not differ (Tukey test, p = 1%).

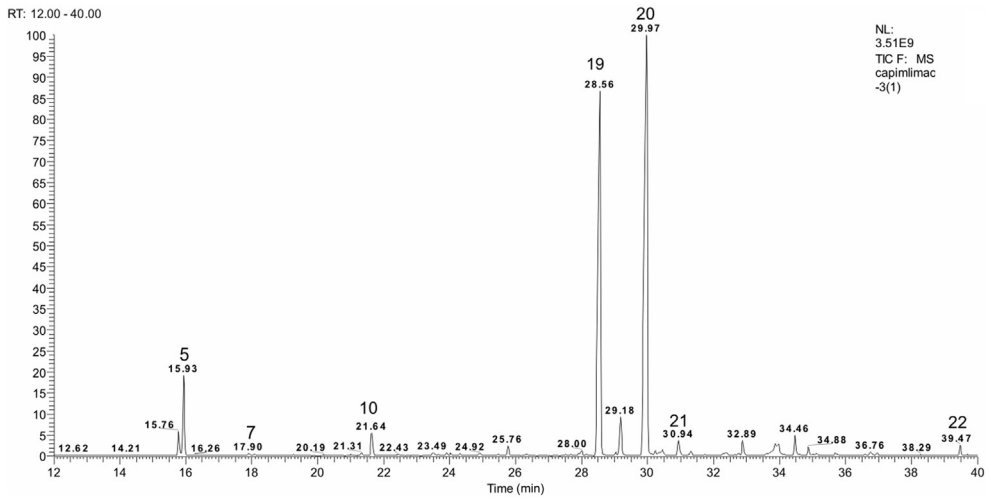


Fig. 6. Expanded chromatographic analysis of essential oil of lemon grass with GC–MS. Compounds: 5) β-myrcene, 7) 1.8-cineol (eucalyptol), 10) linalool, 19) neral, 20) geranial, 21) 2-undecanone, 22) 2-tridecanone.

Balbi-Peña et al. (2012) studied the defensive responses of a genotype of *Solanum habrochaites* resistant to *Oidium neolycopersici* and a genotype of *Solanum lycopersicum* susceptible to this pathogen. Those authors observed an increase in catalase activity 120 h after inoculation, suggesting that detoxification mechanisms ("scavenging" of excess hydrogen peroxide) can act in the later stages of pathogenesis. A similar increase in catalase was also reported in the same pathosystem by Mlíčková et al. (2004) and Tománková et al. (2006).

The reduction of catalase activity found in the present study may represent evidence for the accumulation of hydrogen peroxide and reactive oxygen species (ROS). This response may contribute to the resistance of plants to pathogens (Pascholati, 2011). According to Resende et al. (2003), hydrogen peroxide is involved in many plant resistance mechanisms, acting to reinforce the cell wall by cross-linking structural proteins or by acting on phenolic compounds to form an effective mechanical barrier.

Trouvelot et al. (2008) used microscopy to investigate the activity of β-1,3-glucan sulfate against downy mildew. That study found that the production of hydrogen peroxide is correlated with the hypersensitive response (HR). The authors noted that an oxidative burst and the generation of ROS are observed in response to infection with pathogens belonging to the class Oomycetes, resulting in cell death at the plant infection site (an HR). As a result, the HR is effective in disease resistance (Naton et al., 1996). As the causal agent of downy mildew can also survive in mycelial form in the living tissues of the host, localized cell death can limit the progress of pathogens into the host (Dixon and Lamb, 1997).

Table 4
Components of the essential oil of lemon grass (*C. citratus*) identified by CG-EM and their contents in the area normalization (%).

Components	t_r	KI	Content (%)
β-Mirceno	15.93	990.06	6.11
1.8 Cineol (eucalyptol)	17.9	1027.29	0.19
Linalool	21.64	1098.66	29.96
Neral	28.56	12221.79	2.87
Geranial	29.97	1237.43	41.42
2-Undecanona	30.94	1248.27	1.01
2-Tridecanona	39.47	1495.83	0.71

t_r: Retention time.
KI: Kovats Retention Index.

The chromatogram obtained from the analysis of the essential oil of lemon grass is shown in Fig. 6. The components identified, together with their retention times (t_r), Kovats retention indices (KI) and the sample contents, are shown in Table 4. The chromatographic analysis identified seven compounds in the essential oil of lemon grass: β-myrcene; 1.8 Cineole (eucalyptol); linalool, neral; geranial; 2-undecanone; and 2-tridecanone (Table 4).

In this study, the essential oil of lemon grass was effective in controlling isariopsis leaf spot and downy mildew in grapevine cv. Isabel and induced resistance to these diseases in the plant. It is probable that citral (an isomeric mixture of neral and geranial) was present in the essential oil. Souza et al. (1991) showed that this component of the essential oil of lemon grass possesses substantial antibacterial and antifungal activity. Moreover, Balbi-Peña et al. (2007) found that the essential oil of *C. citratus* induced resistance in tomato plants against *A. solani*, with the highest peroxidase activity found at 12 and 48 h after inoculation compared with the values for a control treatment with water and for non-inoculated plants.

The essential oil of lemon grass contains citral and is effective in controlling leaf spot and leaf and vine mildew in the field. Most likely, the effectiveness of the essential oil for this purpose is due to the production of resistance to pathogens through the induction of the enzymes chitinase and catalase and of other enzymes that may be involved in the response. The use of the essential oil of lemon grass may represent a promising alternative for viticulture, serving to augment the desirable physicochemical characteristics of the grape and increasing productivity and quality.

4. Conclusions

Doses of the essential oil of lemon grass reduced the AUDPC for isariopsis leaf spot and downy mildew in two consecutive years.

Doses of 1.0 and 2.0 mL L^{-1} of the essential oil reduced the severity of downy mildew disease, measured by the AUDPC, by approximately 45%.

A dose of 2.0 mL L^{-1} of the essential oil increased the number of clusters and productivity by 63% and 48%, respectively, in the first crop and by 92% and 150%, respectively, in the second crop.

In the first harvest, a dose of 2.0 mL L^{-1} of essential oil caused a 100% increase in the maturation index, suggesting an effect prior to

grape maturation. In the second harvest, the treatments did not differ significantly.

The highest percentage increases in chitinase activity, 300% and 181%, were observed at doses of 2.0 mL L^{-1} 24 h after the second spraying and 0.5 mL L^{-1} 48 h after the third spraying, respectively.

A chromatographic analysis showed that the essential oil of lemon grass contained citral (an isomeric mixture of neral and geranial). Due to the presence of citral, the essential oil was effective in disease control and in inducing disease resistance in grapevine cv. Isabel.

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